Lab Eleven – Point Estimation and Resampling

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good `ggplot` practice AND will help make sure you don't make a mistake when computing the probability.

1 Introduction

When performing point estimation, there isn't just one *right* answer. For example, the Method of Moments (MOM) and Maximum Likelihood Estimates (MLEs) aren't always the same.

One question that comes up, is why we calculate the sample variance as

$$S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

when it is supposed to be the "average squared distance from the mean, i.e.,

$$S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

2 Comparing Estimators

2.1 Initial Simulation

In this lab, we will explore why we use S_{n-1}^2 and not S_n^2. To do this, we will generate data from four known distributions:

- Poisson$(\lambda = 2)$
- Binomial$\left(n = 50, p = \frac{5-\sqrt{21}}{10}\right)$
- Normal$(\mu = 0, \sigma = \sqrt{2})$
- Exponential$\left(\lambda = \frac{1}{\sqrt{2}}\right)$.

Remark: Note that I have selected the parameters so that the true variance for all four models is 2.

For each distribution, write a loop that completes the following 1000 times. Put `set.seed(7272)` at the top of your code chunk.

- Generate a sample of size n from the distribution.
- Calculate and save S_{n-1}^2 and S_n^2.

Start with $n = 20$. Is there evidence that S_{n-1}^2 is a better estimator than S_n^2?

Solution

```r
set.seed(7272)

#Poisson Distribution
  psv = c() #empty vector to hold sample variance
  pad = c() #empty vector to hold avg. distance from mean
for(i in 1:1000){
  x = rpois(20, 2) #generates sample
  psv[i] = var(x)
  pad[i] = ((20-1)/20)*var(x)
}
#for 1000 simulations, calculuate the sample variance and average squared distance from mean
mean(psv)
```

```
[1] 1.983863
```

```r
mean(pad)
```

```
[1] 1.88467
```

```r
#Binomial Distribution
  bsv = c()
  bad = c()
for(i in 1:1000){
  x = rbinom(20, 50, (5-sqrt(21))/10)
  bsv[i] = var(x)
  bad[i] = ((20-1)/20)*var(x)
}
mean(psv)
```

```
[1] 1.983863
```

```r
mean(pad)
```

```
[1] 1.88467
```

```r
#Normal Distribution
  nsv = c()
  nad = c()
for(i in 1:1000){
  x = rnorm(20, 0, sqrt(2))
  nsv[i] = var(x)
  nad[i] = ((20-1)/20)*var(x)
}
mean(nsv)
```

```
[1] 2.00228
```

```r
mean(nad)
```

```
[1] 1.902166
```

```r
#Exponential Distribution
  esv = c()
  ead = c()
for(i in 1:1000){
  x = rexp(20, 1/(sqrt(2)))
  esv[i] = var(x)
  ead[i] = ((20-1)/20)*var(x)
}
mean(esv)
```

```
[1] 1.984202
```

```r
mean(ead)
```

```
[1] 1.884992
```

There is evidence that S^2_{n-1} is a better estimator than S^2_n as S^2_{n-1} is closer to the true variance of two by about 10%.

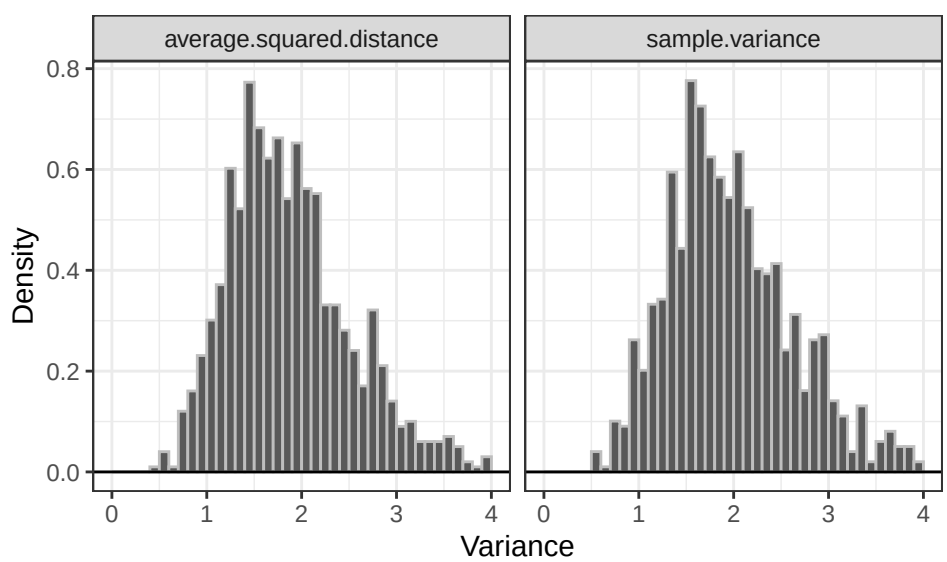
Create a plot or combination of plots (using patchwork) to show the results across all four distributions. Make sure to create a corresponding table that numerically summarizes the results of the simulation. You should include

$$\text{Bias} = \text{Average of Estimates} - \text{True Value}$$
$$\text{Precision} = \frac{1}{\text{Variance of Estimates}}$$
$$\text{Mean Square Error} = \text{Bias}^2 + \text{Variance of Estimates.}$$

Solution

```r
ggdat.p <- tibble(sample.variance = psv,
                  average.squared.distance = pad) |>
 pivot_longer(everything(),
 names_to = "Estimator",
 values_to = "Estimate")
 #create a tibble with columns psv and pad,
 # name the columns estimator
 #call the values estimates

#For a histogram
#breaks fit the possible values of the data
ggplot(ggdat.p) +
 geom_histogram(aes(x = Estimate,
                    y = after_stat(density)),
                    breaks=seq(0,4,0.1), color="grey")+
 geom_hline(yintercept=0)+
 theme_bw() +
 labs(x= "Variance",
      y="Density") +
 facet_wrap(~Estimator)
```

Figure 1: Poisson Variance Estimation.



```r
P.Estimate.Summary = ggdat.p|>
  group_by(Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
#creates a numerical summary of data based on the past distribution
```

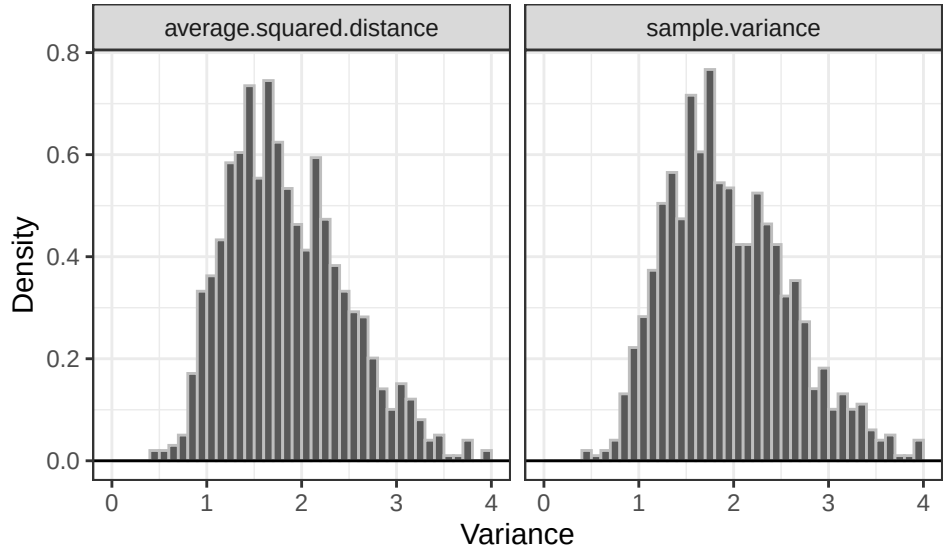
```r
P.Estimate.Summary |>
  knitr::kable(digits = 2)
#creates a table to display summary
```

Table 1: Poisson-Variance-Estimates.

Estimator	Bias	Mean	Precision	MSE
average.squared.distance	-0.12	1.88	2.33	0.44
sample.variance	-0.02	1.98	2.10	0.48

```r
ggdat.b <- tibble(sample.variance = bsv,
                average.squared.distance = bad) |>
 pivot_longer(everything(),
 names_to = "Estimator",
 values_to = "Estimate")
 #create a tibble with columns psv and pad,
 # name the columns estimator
 #call the values estimates

#For a histogram
#breaks fit the possible values of the data
ggplot(ggdat.b) +
 geom_histogram(aes(x = Estimate,
                    y = after_stat(density)),
                    breaks=seq(0,4,0.1), color="grey")+
 geom_hline(yintercept=0)+
 theme_bw() +
 labs(x= "Variance",
      y="Density") +
 facet_wrap(~Estimator)
```

Figure 2: Binomial Variance Estimation.




```
B.Estimate.Summary = ggdat.b|>
  group_by(Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates a numerical summary of data based on the past distribution
```

```
B.Estimate.Summary |>
  knitr::kable(digits = 2)
#creates a table to display summary
```

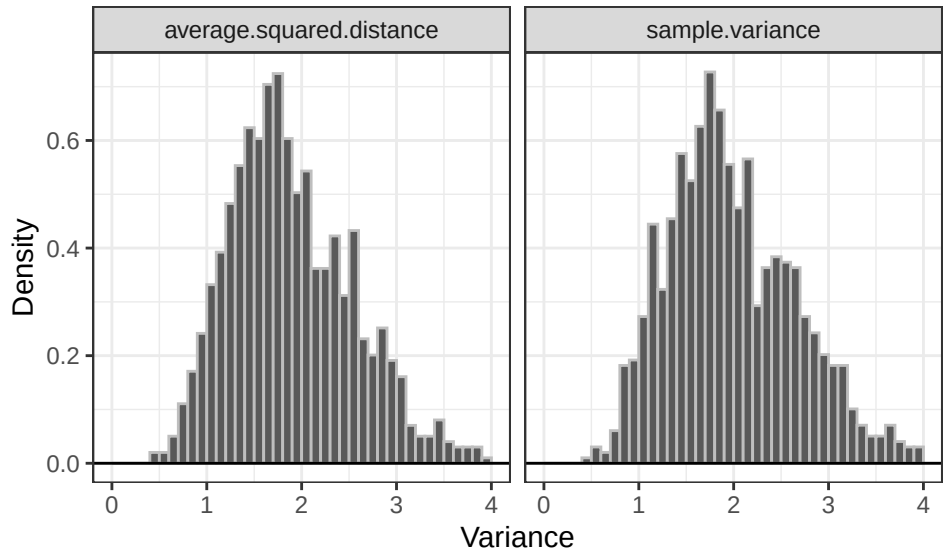
Table 2: Binomial-Variance-Estimates.

Estimator	Bias	Mean	Precision	MSE
average.squared.distance	-0.13	1.87	2.38	0.44
sample.variance	-0.04	1.96	2.15	0.47

```
ggdat.n <- tibble(sample.variance = nsv,
                  average.squared.distance = nad) |>
 pivot_longer(everything(),
 names_to = "Estimator",
 values_to = "Estimate")
 #create a tibble with columns psv and pad,
 # name the columns estimator
 #call the values estimates

#For a histogram
#breaks fit the possible values of the data
ggplot(ggdat.n) +
 geom_histogram(aes(x = Estimate,
                    y = after_stat(density)),
```

```
                    breaks=seq(0,4,0.1), color="grey")+
 geom_hline(yintercept=0)+
 theme_bw() +
 labs(x= "Variance",
      y="Density") +
 facet_wrap(~Estimator)
```

Figure 3: Normal Variance Estimation.




```
N.Estimate.Summary = ggdat.n|>
  group_by(Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates a numerical summary of data based on the past distribution
```

```
N.Estimate.Summary |>
  knitr::kable(digits = 2)
#creates a table to display summary
```

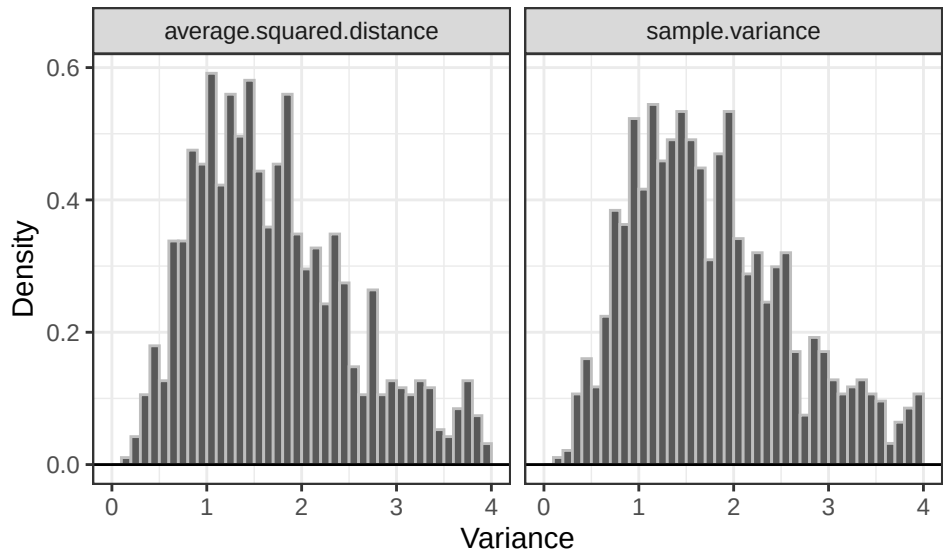
Table 3: Normal-Variance-Estimates.

Estimator	Bias	Mean	Precision	MSE
average.squared.distance	-0.1	1.9	2.25	0.45
sample.variance	0.0	2.0	2.03	0.49

```
ggdat.e <- tibble(sample.variance = esv,
                  average.squared.distance = ead) |>
 pivot_longer(everything(),
 names_to = "Estimator",
 values_to = "Estimate")
 #create a tibble with columns psv and pad,
 # name the columns estimator
```

```r
 #call the values estimates

#For a histogram
#breaks fit the possible values of the data
ggplot(ggdat.e) +
 geom_histogram(aes(x = Estimate,
                    y = after_stat(density)),
                    breaks=seq(0,4,0.1), color="grey")+
 geom_hline(yintercept=0)+
 theme_bw() +
 labs(x= "Variance",
      y="Density") +
 facet_wrap(~Estimator)
```

Figure 4: Exponential Variance Estimation.




```r
E.Estimate.Summary = ggdat.e|>
  group_by(Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates a numerical summary of data based on the past distribution
```

```r
E.Estimate.Summary |>
  knitr::kable(digits = 2)
#creates a table to display summary
```

Table 4: Exponential-Variance-Estimates.

Estimator	Bias	Mean	Precision	MSE
average.squared.distance	-0.12	1.88	0.77	1.30
sample.variance	-0.02	1.98	0.70	1.43

2.2 Simulation over various sample sizes

Consider the effect of the sample size have on these results. Provide a graphical and numerical summary of what happens as n increases – say $n = 20$, $n = 50$, $n = 100$, and $n = 500$. Does the result in your initial simulation hold across samples? Or, do things change as the sample size changes? Ensure to set your randomization seed.

Solution

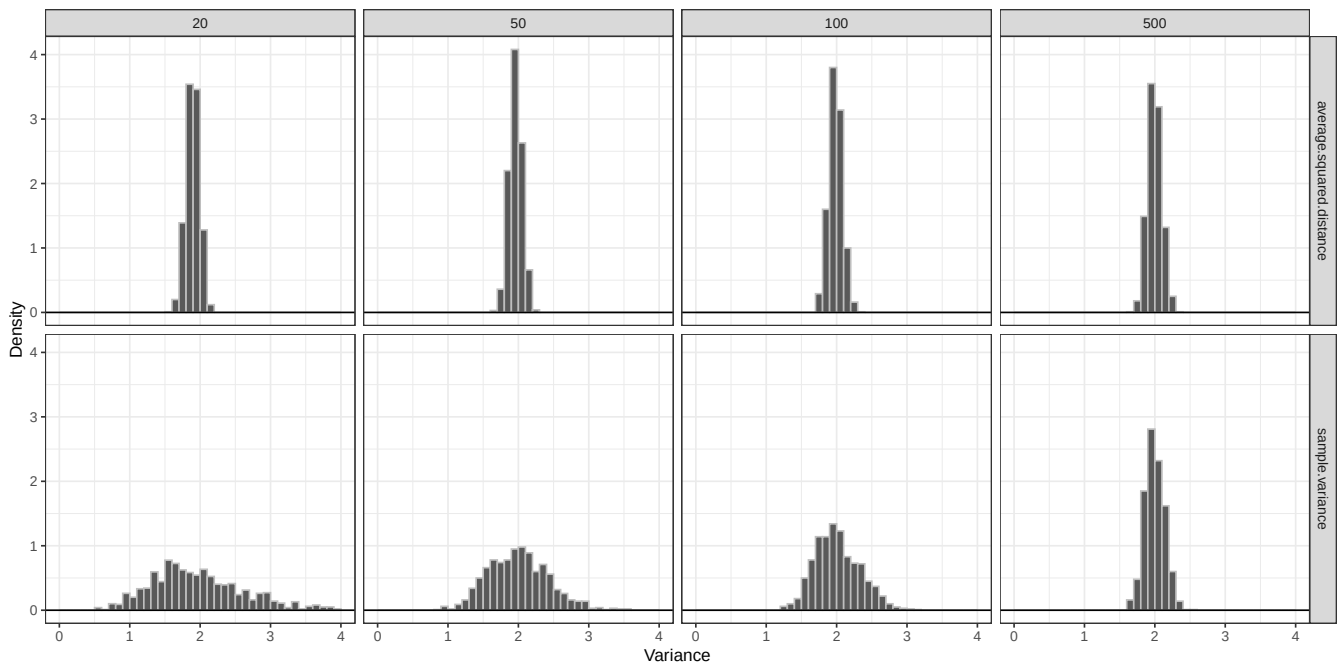
```r
set.seed(7272)

#Poisson Distribution
  psv = function(n) {
    psv_vals = numeric(1000)
    for (i in 1:1000) {
      x = rpois(n, 2)
      psv_vals[i] = var(x)
    }

    return(psv_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
  pad = function(n) {
    pad_vals = numeric(1000)
    for (i in 1:1000) {
      x = rpois(n, 2)
      pad_vals[i] = ((n-1)/n)*var(x)
    }

    return(pad_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
#Binomial Distribution
  bsv = function(n) {
    bsv_vals = numeric(1000)
    for (i in 1:1000) {
      x = rbinom(n, 50, (5-sqrt(21))/10)
      bsv_vals[i] = var(x)
    }

    return(bsv_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
  bad = function(n) {
    bad_vals = numeric(1000)
    for (i in 1:1000) {
      x = rbinom(n, 50, (5-sqrt(21))/10)
      bad_vals[i] = ((n-1)/n)*var(x)
    }

    return(bad_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
#Normal Distribution

  nsv = function(n) {
    nsv_vals = numeric(1000)
    for (i in 1:1000) {
      x =  rnorm(n, 0, sqrt(2))
```

```r
      nsv_vals[i] = var(x)
    }

    return(nsv_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
  nad = function(n) {
    nad_vals = numeric(1000)
    for (i in 1:1000) {
      x =  rnorm(n, 0, sqrt(2))
      nad_vals[i] = ((n-1)/n)*var(x)
    }

    return(nad_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
#Exponential Distribution
  esv = function(n) {
    esv_vals = numeric(1000)
    for (i in 1:1000) {
      x =  rexp(n, 1/(sqrt(2)))
      esv_vals[i] = var(x)
    }

    return(esv_vals)
  }
#creates function in terms of n that calculates 1000 simulations of size n
  ead = function(n) {
    ead_vals = numeric(1000)
    for (i in 1:1000) {
      x =  rexp(n, 1/(sqrt(2)))
      ead_vals[i] = ((n-1)/n)*var(x)
    }

    return(ead_vals)
  }
  #creates function in terms of n that calculates 1000 simulations of size n
```

```r
n_values = c(20, 50, 100, 500)
#creates vector to pass through function

ggdat.p <- lapply(n_values, function(n) {
  tibble(
    n = n,
    sample.variance = psv(n),
    average.squared.distance = pad(n)
  )
}) |>
  bind_rows() |>
  pivot_longer(
    cols = c(sample.variance, average.squared.distance),
    names_to = "Estimator",
    values_to = "Estimate"
  )
#creates data set in terms of the n vector with columns for estimates

#creates histogram faceted on both the estimator and n value
```

```r
ggplot(ggdat.p) +
  geom_histogram(
    aes(x = Estimate, y = after_stat(density)),
    breaks = seq(0, 4, 0.1),
    color = "grey"
  ) +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance", y = "Density") +
  facet_grid(Estimator ~ n)
```

Figure 5: Poisson Variance Estimation.




```r
P.Estimate.Summary = ggdat.p|>
  group_by(n, Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Variance = var(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates data summary grouping by both n and estimator
```

```r
P.Estimate.Summary |>
  knitr::kable(digits = 2)
```

Table 5: Poisson-Variance-Estimates.

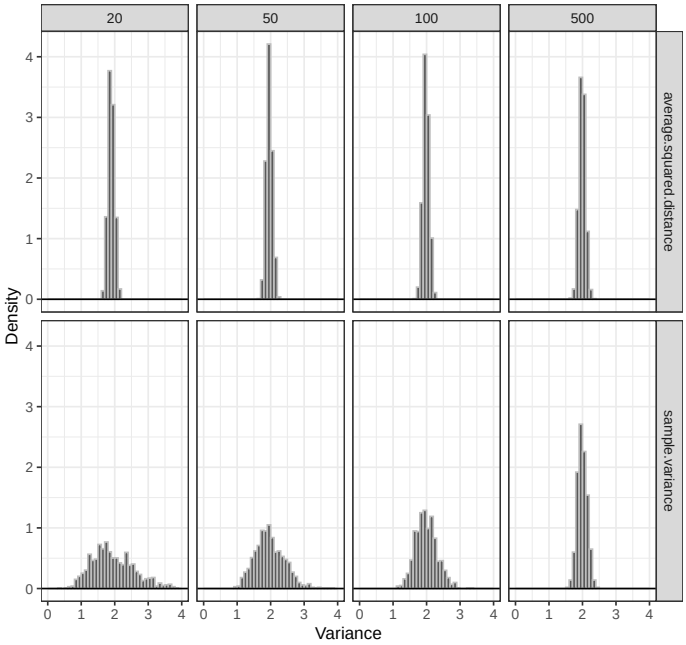
n	Estimator	Bias	Mean	Variance	Precision	MSE
20	average.squared.distance	-0.10	1.90	0.01	112.84	0.02
20	sample.variance	-0.02	1.98	0.48	2.10	0.48
50	average.squared.distance	-0.04	1.96	0.01	115.19	0.01

Table 5: Poisson-Variance-Estimates.

n	Estimator	Bias	Mean	Variance	Precision	MSE
50	sample.variance	0.00	2.00	0.19	5.39	0.19
100	average.squared.distance	-0.02	1.98	0.01	108.72	0.01
100	sample.variance	0.01	2.01	0.10	9.72	0.10
500	average.squared.distance	0.00	2.00	0.01	99.13	0.01
500	sample.variance	0.00	2.00	0.02	52.74	0.02

```r
n_values = c(20, 50, 100, 500)
#creates vector to pass through function

ggdat.b = lapply(n_values, function(n) {
  tibble(
    n = n,
    sample.variance = bsv(n),
    average.squared.distance = bad(n)
  )
}) |>
  bind_rows() |>
  pivot_longer(
    cols = c(sample.variance, average.squared.distance),
    names_to = "Estimator",
    values_to = "Estimate"
  )
#creates data set in terms of the n vector with columns for estimates

#creates histogram faceted on both the estimator and n value
ggplot(ggdat.p) +
ggplot(ggdat.b) +
  geom_histogram(
    aes(x = Estimate, y = after_stat(density)),
    breaks = seq(0, 4, 0.1),
    color = "grey"
  ) +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance", y = "Density") +
  facet_grid(Estimator ~ n)
```

Figure 6: Binomial Variance Estimation.




```r
B.Estimate.Summary = ggdat.b|>
  group_by(n, Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Variance = var(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates data summary grouping by both n and estimator
```

```r
B.Estimate.Summary |>
  knitr::kable(digits = 2)
```

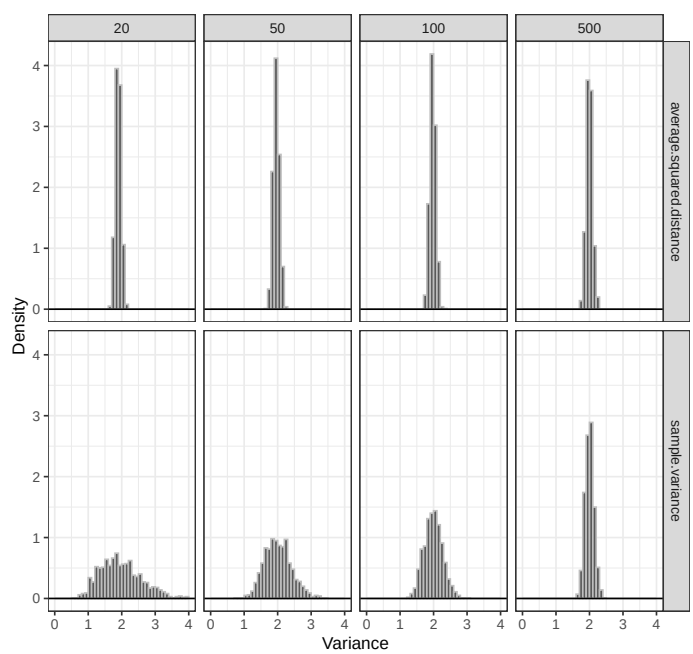
Table 6: Binomial-Variance-Estimates.

n	Estimator	Bias	Mean	Variance	Precision	MSE
20	average.squared.distance	-0.10	1.90	0.01	111.01	0.02
20	sample.variance	-0.03	1.97	0.46	2.15	0.47
50	average.squared.distance	-0.04	1.96	0.01	120.99	0.01
50	sample.variance	-0.02	1.98	0.20	5.02	0.20
100	average.squared.distance	-0.02	1.98	0.01	112.35	0.01
100	sample.variance	-0.01	1.99	0.10	9.56	0.10
500	average.squared.distance	-0.01	1.99	0.01	107.63	0.01
500	sample.variance	-0.01	1.99	0.02	50.20	0.02

```r
n_values = c(20, 50, 100, 500)
#creates vector to pass through function

ggdat.n = lapply(n_values, function(n) {
  tibble(
```

```r
    n = n,
    sample.variance = nsv(n),
    average.squared.distance = nad(n)
  )
}) |>
  bind_rows() |>
  pivot_longer(
    cols = c(sample.variance, average.squared.distance),
    names_to = "Estimator",
    values_to = "Estimate"
  )
#creates data set in terms of the n vector with columns for estimates

#creates histogram faceted on both the estimator and n value
ggplot(ggdat.p) +
ggplot(ggdat.n) +
  geom_histogram(
    aes(x = Estimate, y = after_stat(density)),
    breaks = seq(0, 4, 0.1),
    color = "grey"
  ) +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance", y = "Density") +
  facet_grid(Estimator ~ n)
```

Figure 7: Normal Variance Estimation.




```r
N.Estimate.Summary = ggdat.n|>
  group_by(n, Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Variance = var(Estimate),
```

```r
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates data summary grouping by both n and estimator
```

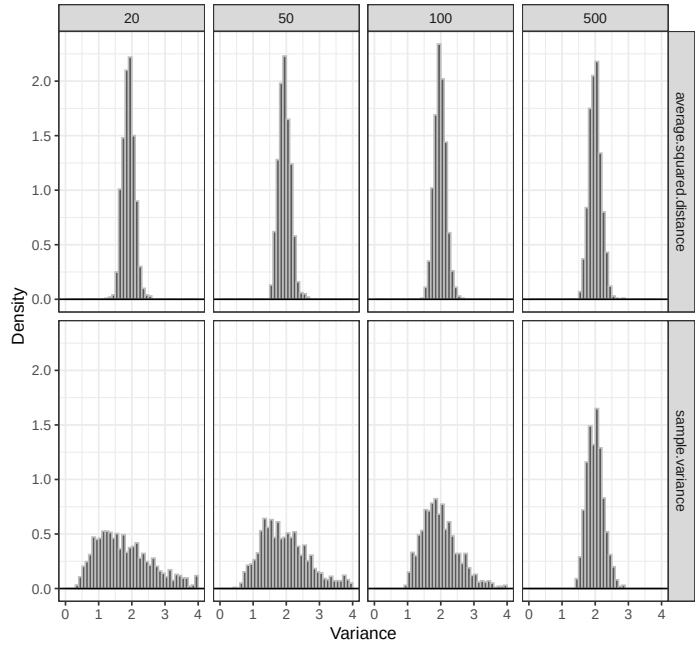
```r
N.Estimate.Summary |>
  knitr::kable(digits = 2)
```

Table 7: Normal-Variance-Estimates.

n	Estimator	Bias	Mean	Variance	Precision	MSE
20	average.squared.distance	-0.10	1.90	0.01	141.04	0.02
20	sample.variance	-0.02	1.98	0.40	2.48	0.40
50	average.squared.distance	-0.04	1.96	0.01	116.30	0.01
50	sample.variance	0.01	2.01	0.17	5.96	0.17
100	average.squared.distance	-0.02	1.98	0.01	128.50	0.01
100	sample.variance	0.00	2.00	0.08	13.27	0.08
500	average.squared.distance	0.00	2.00	0.01	118.10	0.01
500	sample.variance	0.00	2.00	0.02	59.21	0.02

```r
n_values = c(20, 50, 100, 500)
#creates vector to pass through function

ggdat.e = lapply(n_values, function(n) {
  tibble(
    n = n,
    sample.variance = esv(n),
    average.squared.distance = ead(n)
  )
}) |>
  bind_rows() |>
  pivot_longer(
    cols = c(sample.variance, average.squared.distance),
    names_to = "Estimator",
    values_to = "Estimate"
  )
#creates data set in terms of the n vector with columns for estimates

#creates histogram faceted on both the estimator and n value
ggplot(ggdat.p) +
ggplot(ggdat.e) +
  geom_histogram(
    aes(x = Estimate, y = after_stat(density)),
    breaks = seq(0, 4, 0.1),
    color = "grey"
  ) +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance", y = "Density") +
  facet_grid(Estimator ~ n)
```

Figure 8: Exponential Variance Estimation.




```r
E.Estimate.Summary = ggdat.e|>
  group_by(n, Estimator) |>
  summarize(
    Bias = mean(Estimate) - 2,
    Mean = mean(Estimate),
    Variance = var(Estimate),
    Precision = 1/(var(Estimate)),
    MSE = Bias^2 + var(Estimate),
  )
  #creates data summary grouping by both n and estimator
```

```r
E.Estimate.Summary |>
  knitr::kable(digits = 2)
```

Table 8: Exponential-Variance-Estimates.

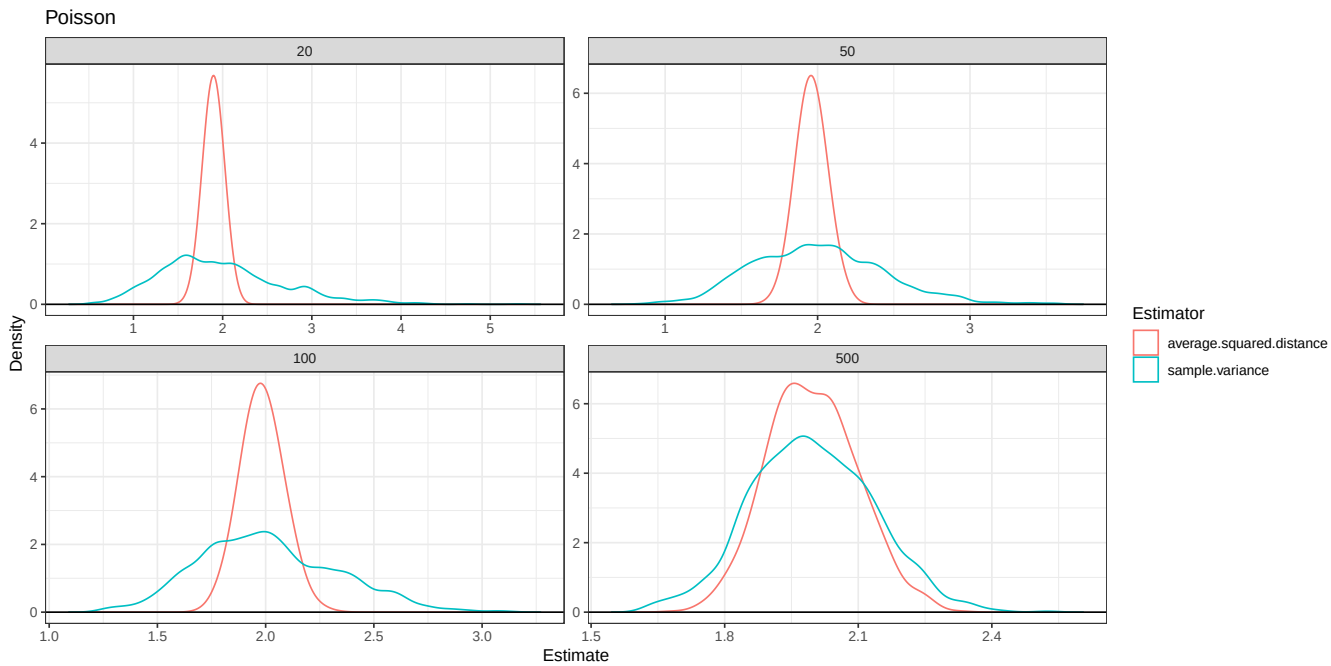
n	Estimator	Bias	Mean	Variance	Precision	MSE
20	average.squared.distance	-0.09	1.91	0.03	31.74	0.04
20	sample.variance	0.01	2.01	1.42	0.70	1.42
50	average.squared.distance	-0.05	1.95	0.03	30.74	0.03
50	sample.variance	0.01	2.01	0.66	1.52	0.66
100	average.squared.distance	-0.02	1.98	0.03	32.10	0.03
100	sample.variance	0.00	2.00	0.33	3.06	0.33
500	average.squared.distance	0.00	2.00	0.03	29.66	0.03
500	sample.variance	0.00	2.00	0.06	16.24	0.06

2.3 Simulation

You have written code to draw a random sample of various sizes from each distribution, and to calculate and store the calculated estimates. Rework your code to produce a density plot of the s_{n-1}^2 and s_n^2. Do you have a guess about their distribution? Ensure to set your randomization seed.

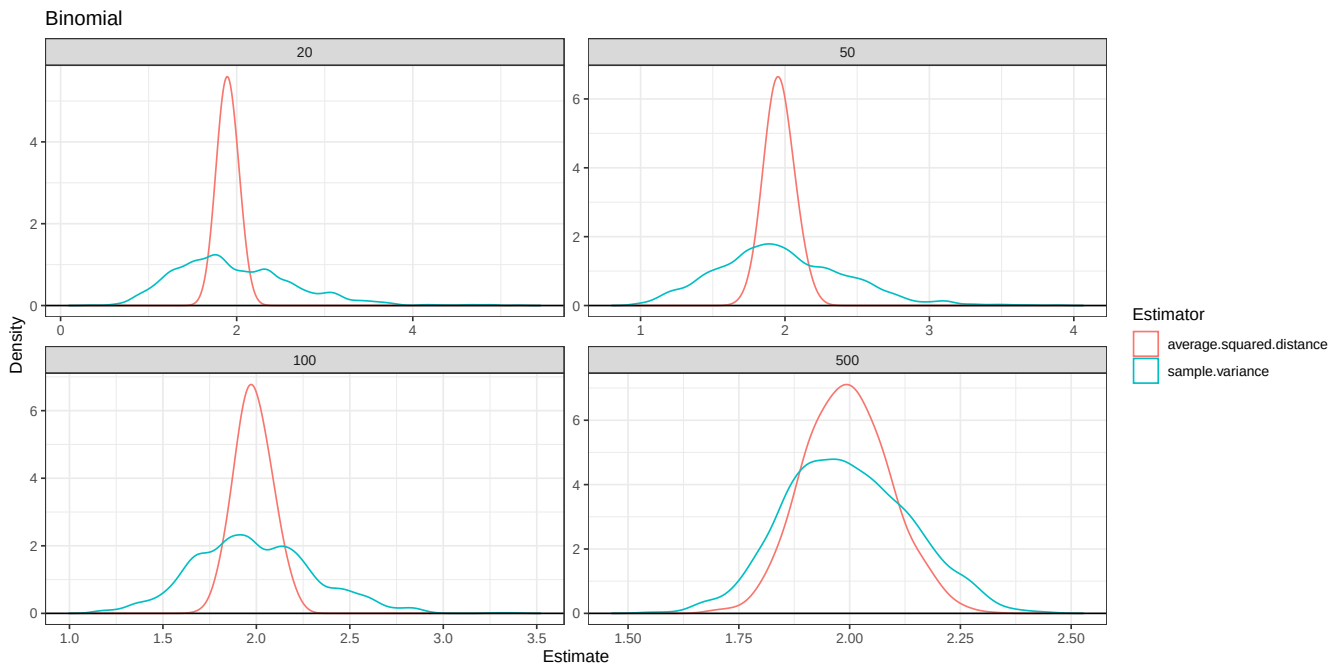
```r
set.seed(7272)
library(ggridges)

ggplot(ggdat.p) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Poisson") +
facet_wrap(~ n, scales = "free") +
# Each facet panel can have it's own scale
ylab("Density") +
theme_bw()
```

Figure 9: Poisson Density Plot Variance Estimation.



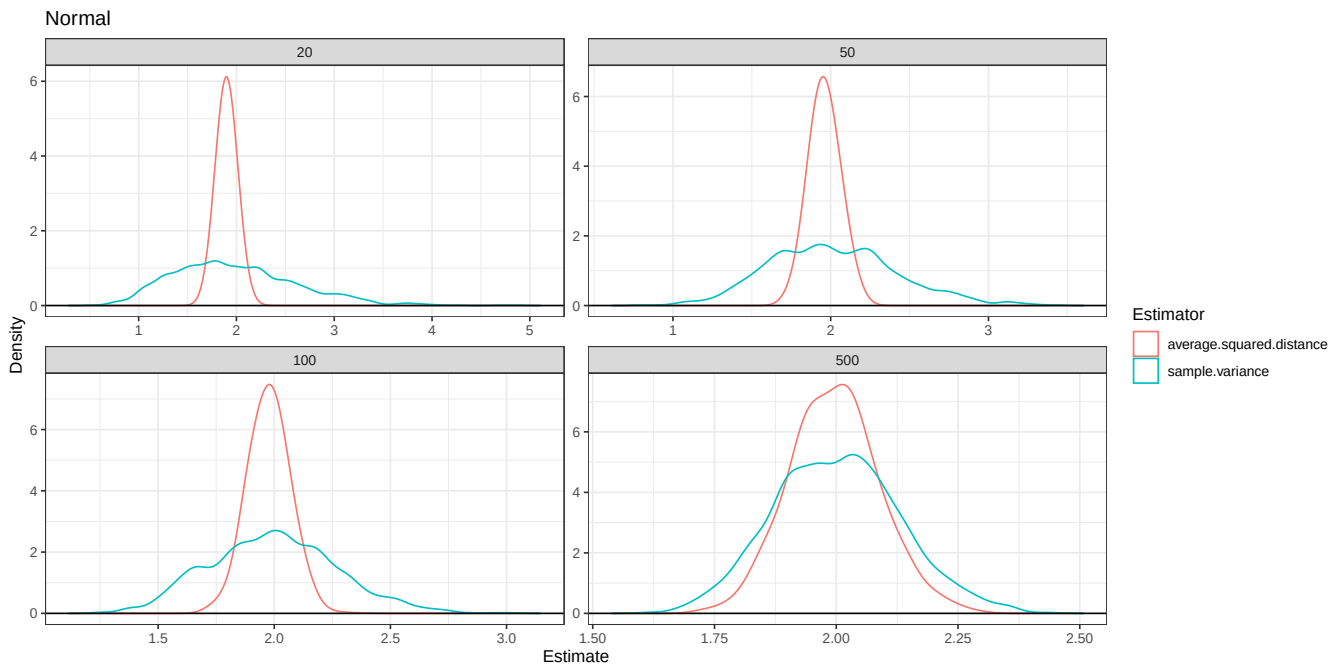

```r
set.seed(7272)

ggplot(ggdat.b) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Binomial") +
facet_wrap(~ n, scales = "free") +
# Each facet panel can have it's own scale
ylab("Density") +
theme_bw()
```

Figure 10: Binomial Density Plot Variance Estimation.



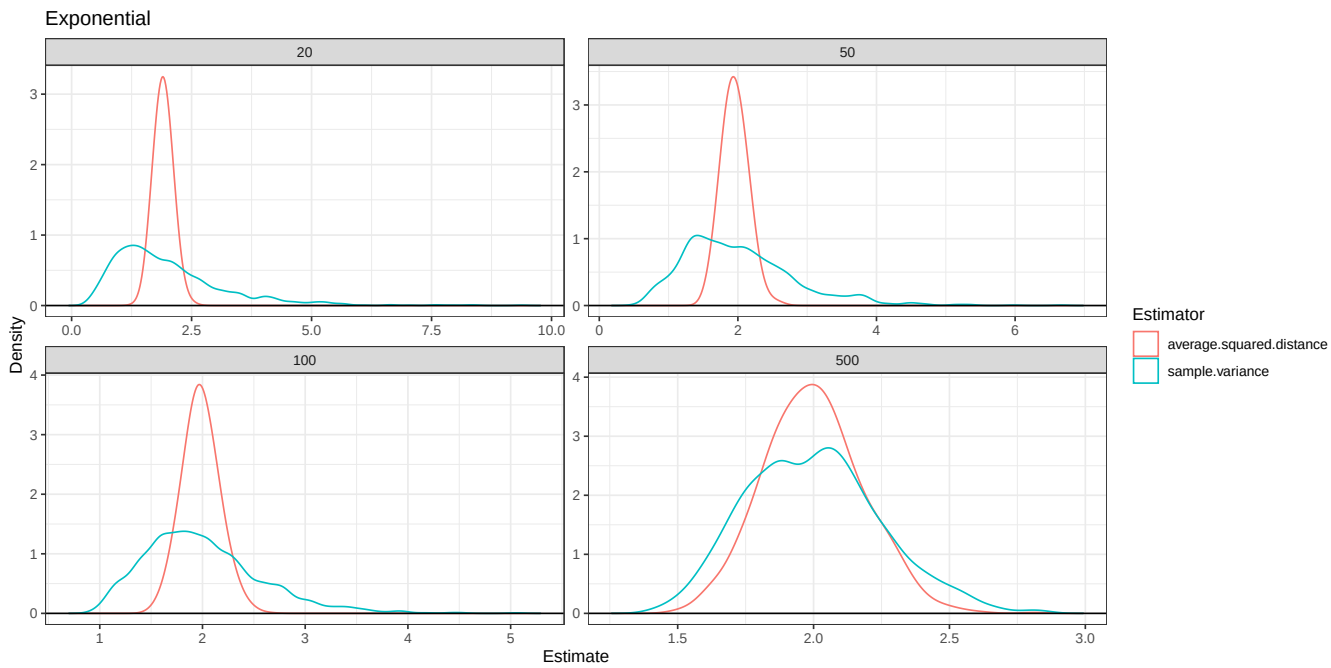

```
set.seed(7272)

ggplot(ggdat.n) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Normal") +
facet_wrap(~ n, scales = "free") +
# Each facet panel can have it's own scale
ylab("Density") +
theme_bw()
```

Figure 11: Normal Density Plot Variance Estimation.




```r
set.seed(7272)

ggplot(ggdat.e) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Exponential") +
facet_wrap(~ n, scales = "free") +
# Each facet panel can have it's own scale
ylab("Density") +
theme_bw()
```

Figure 12: Exponential Density Plot Variance Estimation.



2.4 Bootstrap

Now, take *one* sample from each distribution. Instead of drawing a random sample at each iteration, sample *with replacement* from the one sample. Produce a density plot of the s^2_{n-1} and s^2_n. Do we get roughly the same information as we do in the full simulation? Try this over several seeds – what do you notice? Ensure to set your randomization seed.

Solution

With the bootstrap sample, as with having multiple smaples, the sampling distributions of variance typically follow a Gaussian distribution, becoming more normal and symmetric as the number of replacements increase (similar to if the number of samples increased). However, using bootstrapping can lead to less accurate data, as these distributions were centered at values other than the true population variance of two. Thus, it is important that only a representative sample is bootstrapped, which is often incredibly difficult. Additionally, as the number of replacements increased, the respective distributions of average squared distance away from the mean and the sample variance became very similar.

With Seed(7272)

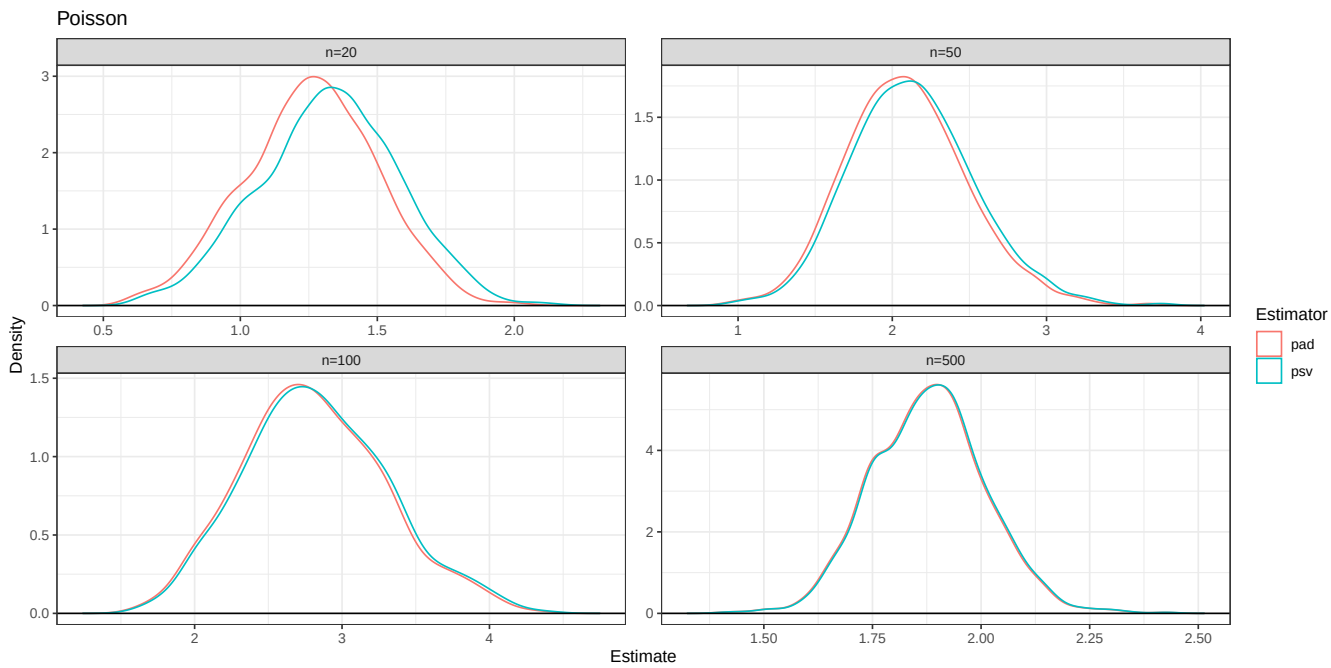
```
set.seed(7272)

n.values = c(20, 50, 100, 500)

# poisson
 pois.sample = list()
 psv_b = list()
 pad_b = list()
 psv = rep(NA, 1000)
 pad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  pois.sample[[as.character(n)]] = rpois(n, 2)
  for(i in 1:1000){
```

```r
    curr.sample = sample(pois.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    psv[i] = sum((curr.sample - x)^2) / (n - 1)
    pad[i] = sum((curr.sample - x)^2) / n
  }
  psv_b[[as.character(n)]] = psv
  pad_b[[as.character(n)]] = pad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# binomial
 binom.sample = list()
 bsv_b = list()
 bad_b = list()
 bsv = rep(NA, 1000)
 bad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  binom.sample[[as.character(n)]] = rbinom(n, 50, (5-sqrt(21))/10)
  for(i in 1:1000){
    curr.sample = sample(binom.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    bsv[i] = sum((curr.sample - x)^2) / (n - 1)
    bad[i] = sum((curr.sample - x)^2) / n
  }
  bsv_b[[as.character(n)]] = bsv
  bad_b[[as.character(n)]] = bad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# normal
 norm.sample = list()
 nsv_b = list()
 nad_b = list()
 nsv = rep(NA, 1000)
 nad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  norm.sample[[as.character(n)]] = rnorm(n, 0, sqrt(2))
  for(i in 1:1000){
    curr.sample = sample(norm.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    nsv[i] = sum((curr.sample - x)^2) / (n - 1)
    nad[i] = sum((curr.sample - x)^2) / n
  }
  nsv_b[[as.character(n)]] = nsv
  nad_b[[as.character(n)]] = nad
}
# creates random sample that is sampled 1000 times
#calculates the estimates
```

```r

# exponential
 exp.sample = list()
 esv_b = list()
 ead_b = list()
 esv = rep(NA, 1000)
 ead = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  exp.sample[[as.character(n)]] = rexp(n, 1/(sqrt(2)))
  for(i in 1:1000){
    curr.sample = sample(exp.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    esv[i] = sum((curr.sample - x)^2) / (n - 1)
    ead[i] = sum((curr.sample - x)^2) / n
  }
  esv_b[[as.character(n)]] = esv
  ead_b[[as.character(n)]] = ead
}
# creates random sample that is sampled 1000 times
#calculates the estimates
```

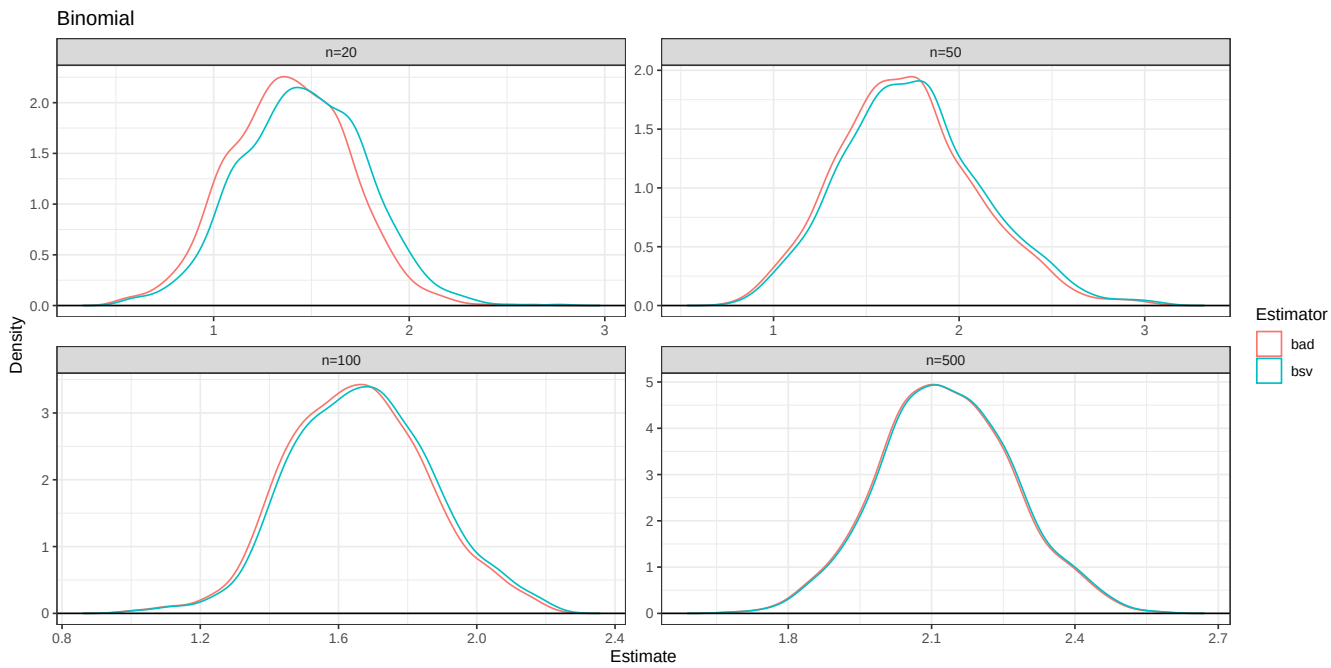
```r
ggdat.pb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(psv_b[["20"]], pad_b[["20"]],
psv_b[["50"]], pad_b[["50"]],
psv_b[["100"]], pad_b[["100"]],
psv_b[["500"]], pad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.pb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Poisson") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 13: Poisson Density Plot Variance Estimation with Bootstrapping.



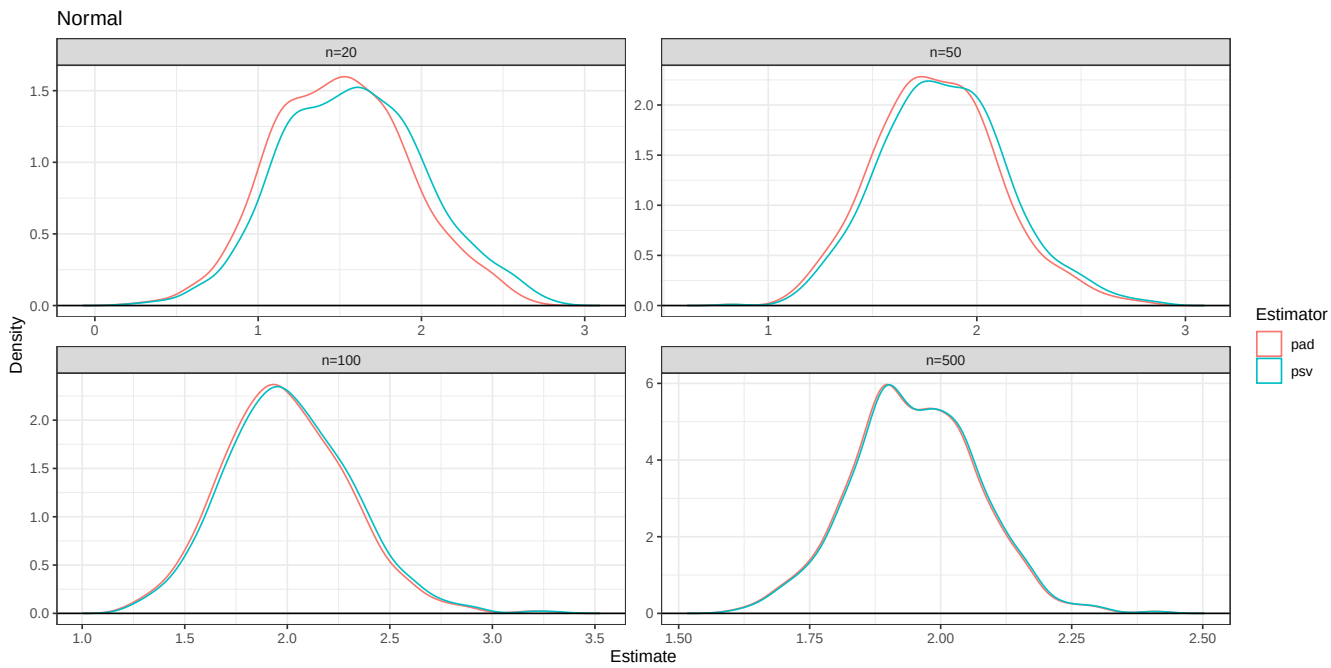

```r
ggdat.bb <- tibble(
Estimator = rep(c("bsv", "bad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(bsv_b[["20"]], bad_b[["20"]],
bsv_b[["50"]], bad_b[["50"]],
bsv_b[["100"]], bad_b[["100"]],
bsv_b[["500"]], bad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.bb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Binomial") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 14: Binomial Density Plot Variance Estimation with Bootstrapping.



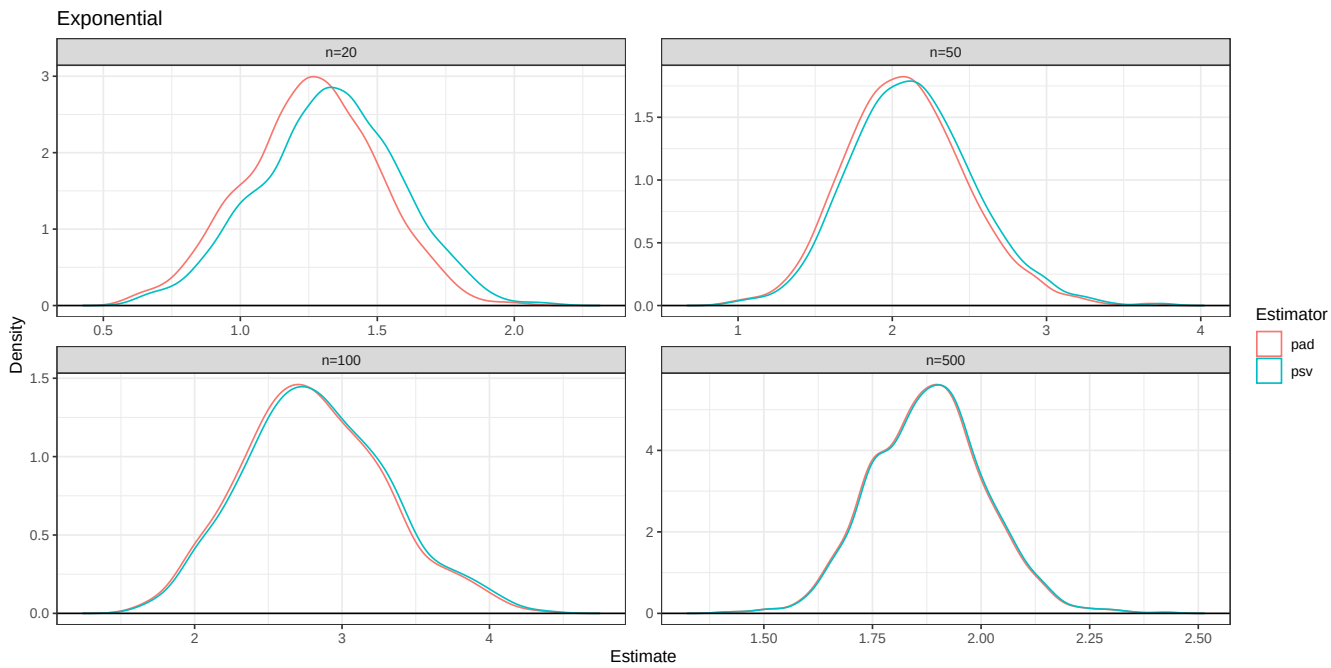

```r
ggdat.nb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(nsv_b[["20"]], nad_b[["20"]],
nsv_b[["50"]], nad_b[["50"]],
nsv_b[["100"]], nad_b[["100"]],
nsv_b[["500"]], nad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.nb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Normal") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 15: Normal Density Plot Variance Estimation with Bootstrapping.




```r
ggdat.eb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(psv_b[["20"]], pad_b[["20"]],
psv_b[["50"]], pad_b[["50"]],
psv_b[["100"]], pad_b[["100"]],
psv_b[["500"]], pad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.eb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Exponential") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 16: Exponential Density Plot Variance Estimation with Bootstrapping.



With seed(1325)

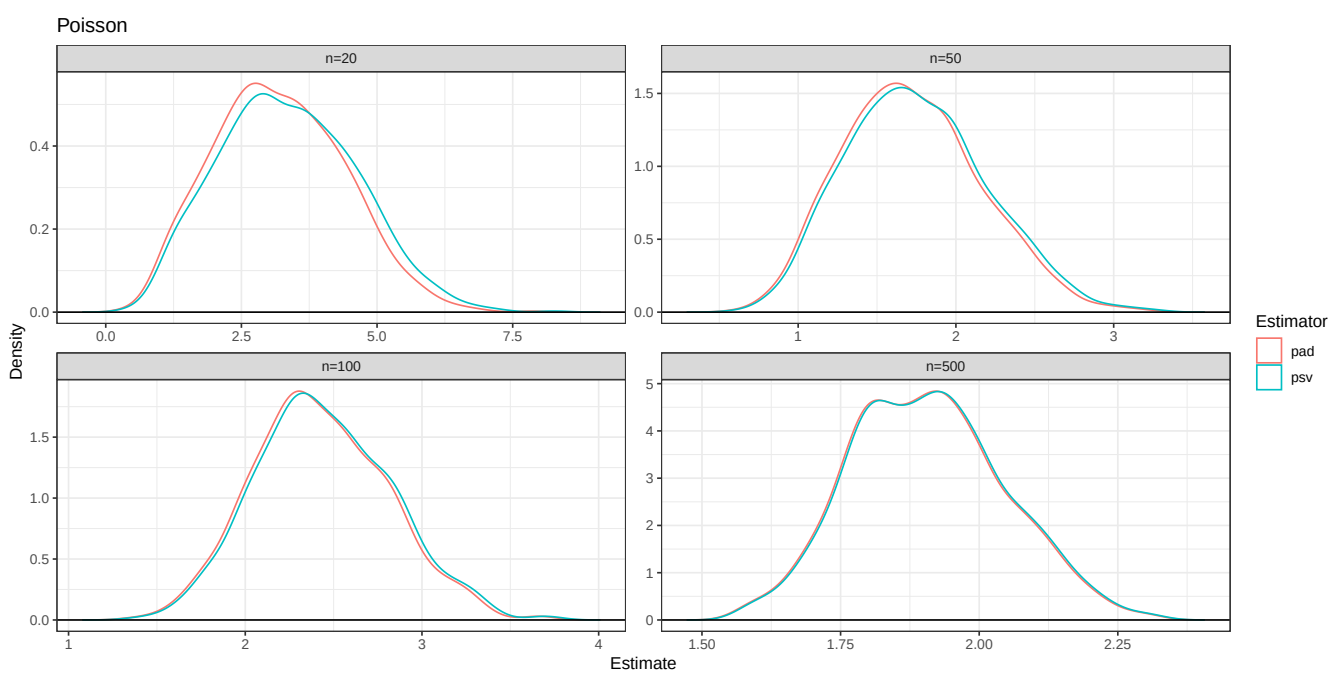
```
set.seed(1325)

n.values = c(20, 50, 100, 500)

# poisson
 pois.sample = list()
 psv_b = list()
 pad_b = list()
 psv = rep(NA, 1000)
 pad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  pois.sample[[as.character(n)]] = rpois(n, 2)
  for(i in 1:1000){
    curr.sample = sample(pois.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    psv[i] = sum((curr.sample - x)^2) / (n - 1)
    pad[i] = sum((curr.sample - x)^2) / n
  }
  psv_b[[as.character(n)]] = psv
  pad_b[[as.character(n)]] = pad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# binomial
 binom.sample = list()
 bsv_b = list()
 bad_b = list()
```

```r
 bsv = rep(NA, 1000)
 bad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  binom.sample[[as.character(n)]] = rbinom(n, 50, (5-sqrt(21))/10)
  for(i in 1:1000){
    curr.sample = sample(binom.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    bsv[i] = sum((curr.sample - x)^2) / (n - 1)
    bad[i] = sum((curr.sample - x)^2) / n
  }
  bsv_b[[as.character(n)]] = bsv
  bad_b[[as.character(n)]] = bad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# normal
 norm.sample = list()
 nsv_b = list()
 nad_b = list()
 nsv = rep(NA, 1000)
 nad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  norm.sample[[as.character(n)]] = rnorm(n, 0, sqrt(2))
  for(i in 1:1000){
    curr.sample = sample(norm.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    nsv[i] = sum((curr.sample - x)^2) / (n - 1)
    nad[i] = sum((curr.sample - x)^2) / n
  }
  nsv_b[[as.character(n)]] = nsv
  nad_b[[as.character(n)]] = nad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# exponential
 exp.sample = list()
 esv_b = list()
 ead_b = list()
 esv = rep(NA, 1000)
 ead = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  exp.sample[[as.character(n)]] = rexp(n, 1/(sqrt(2)))
  for(i in 1:1000){
    curr.sample = sample(exp.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
```

```
    esv[i] = sum((curr.sample - x)^2) / (n - 1)
    ead[i] = sum((curr.sample - x)^2) / n
  }
  esv_b[[as.character(n)]] = esv
  ead_b[[as.character(n)]] = ead
}
# creates random sample that is sampled 1000 times
#calculates the estimates
```

```
ggdat.pb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(psv_b[["20"]], pad_b[["20"]],
psv_b[["50"]], pad_b[["50"]],
psv_b[["100"]], pad_b[["100"]],
psv_b[["500"]], pad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.pb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Poisson") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

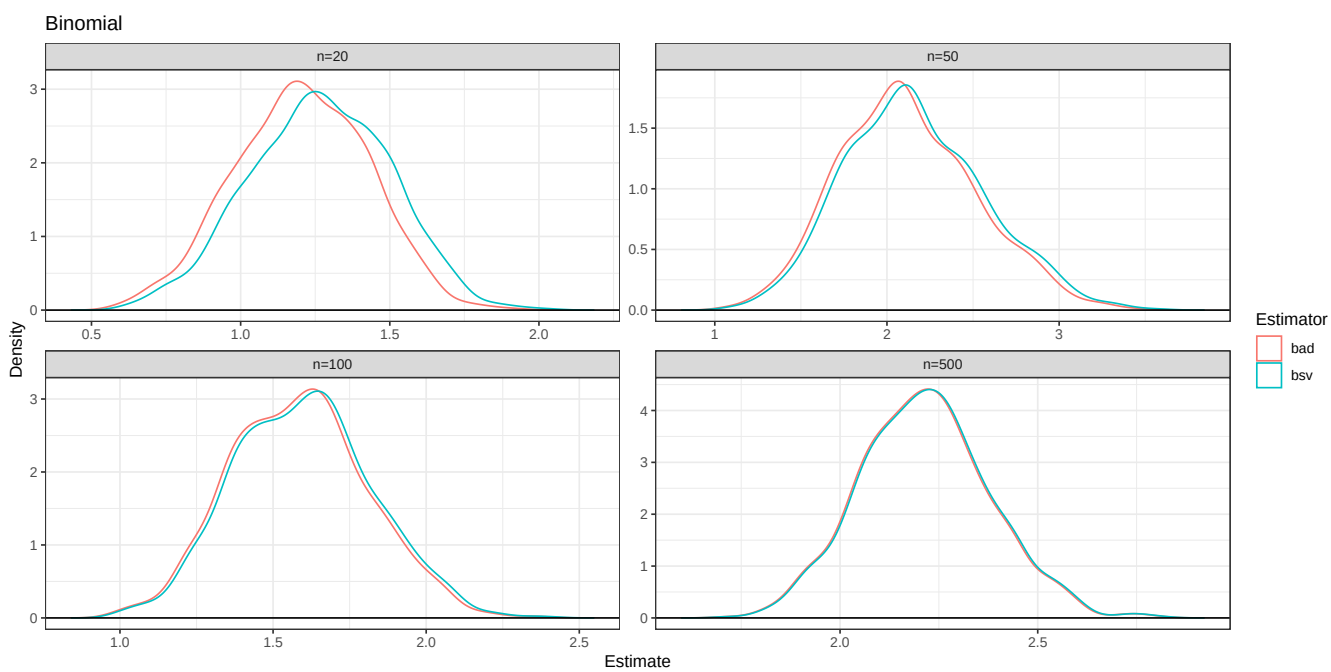
Figure 17: Poisson Density Plot Variance Estimation with Bootstrapping.




```
ggdat.bb <- tibble(
Estimator = rep(c("bsv", "bad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
```

```
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(bsv_b[["20"]], bad_b[["20"]],
bsv_b[["50"]], bad_b[["50"]],
bsv_b[["100"]], bad_b[["100"]],
bsv_b[["500"]], bad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.bb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Binomial") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

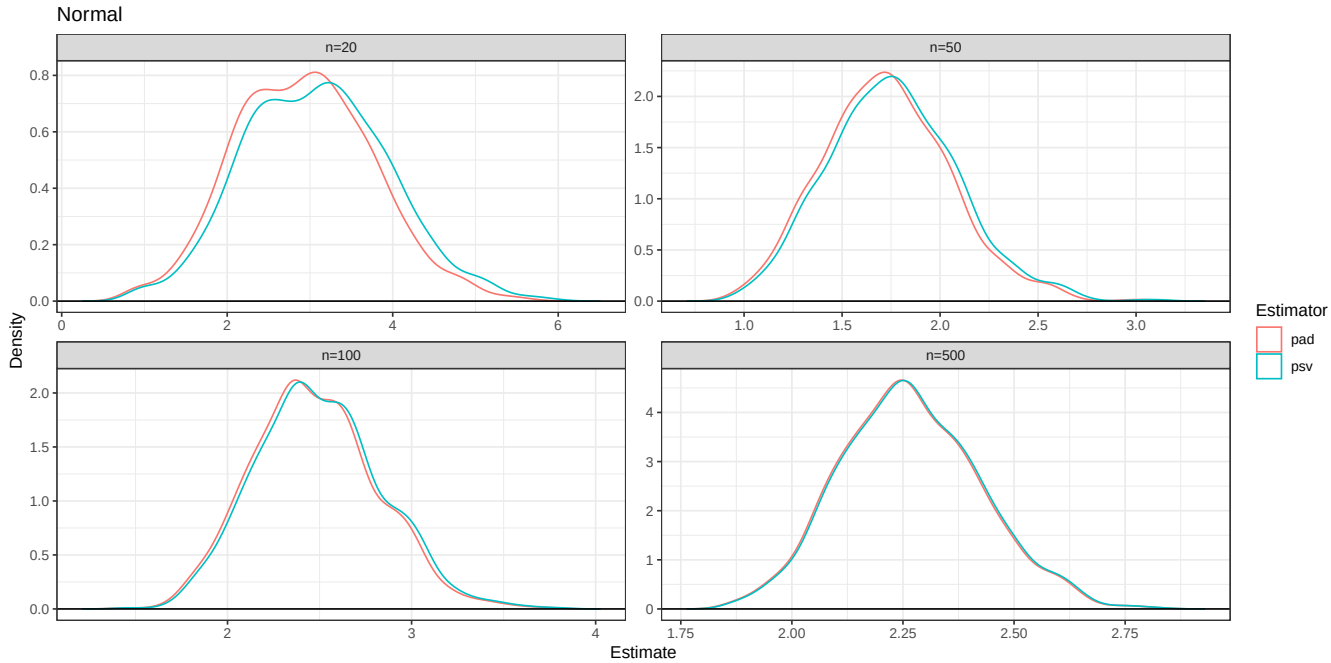
Figure 18: Binomial Density Plot Variance Estimation with Bootstrapping.




```
ggdat.nb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(nsv_b[["20"]], nad_b[["20"]],
nsv_b[["50"]], nad_b[["50"]],
nsv_b[["100"]], nad_b[["100"]],
nsv_b[["500"]], nad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.nb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Normal") +
```

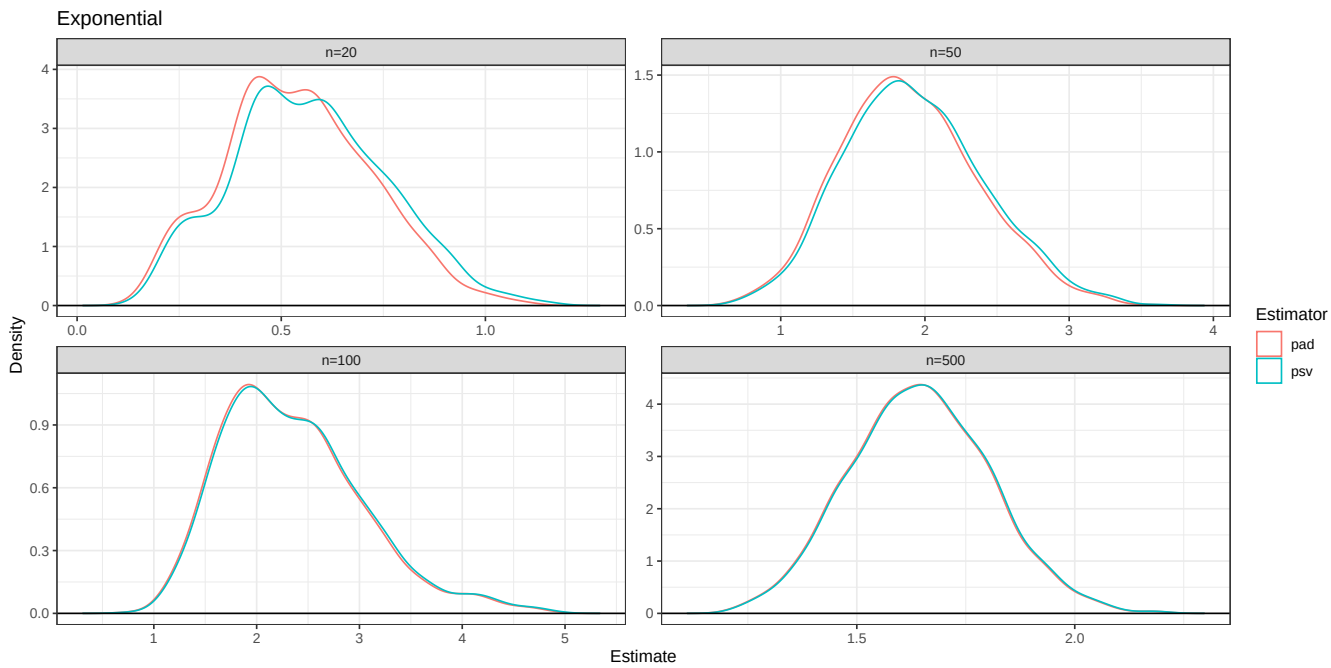
```r
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 19: Normal Density Plot Variance Estimation with Bootstrapping.




```r
ggdat.eb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(esv_b[["20"]], ead_b[["20"]],
esv_b[["50"]], ead_b[["50"]],
esv_b[["100"]], ead_b[["100"]],
esv_b[["500"]], ead_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.eb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Exponential") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 20: Exponential Density Plot Variance Estimation with Bootstrapping.



With Seed(3434)

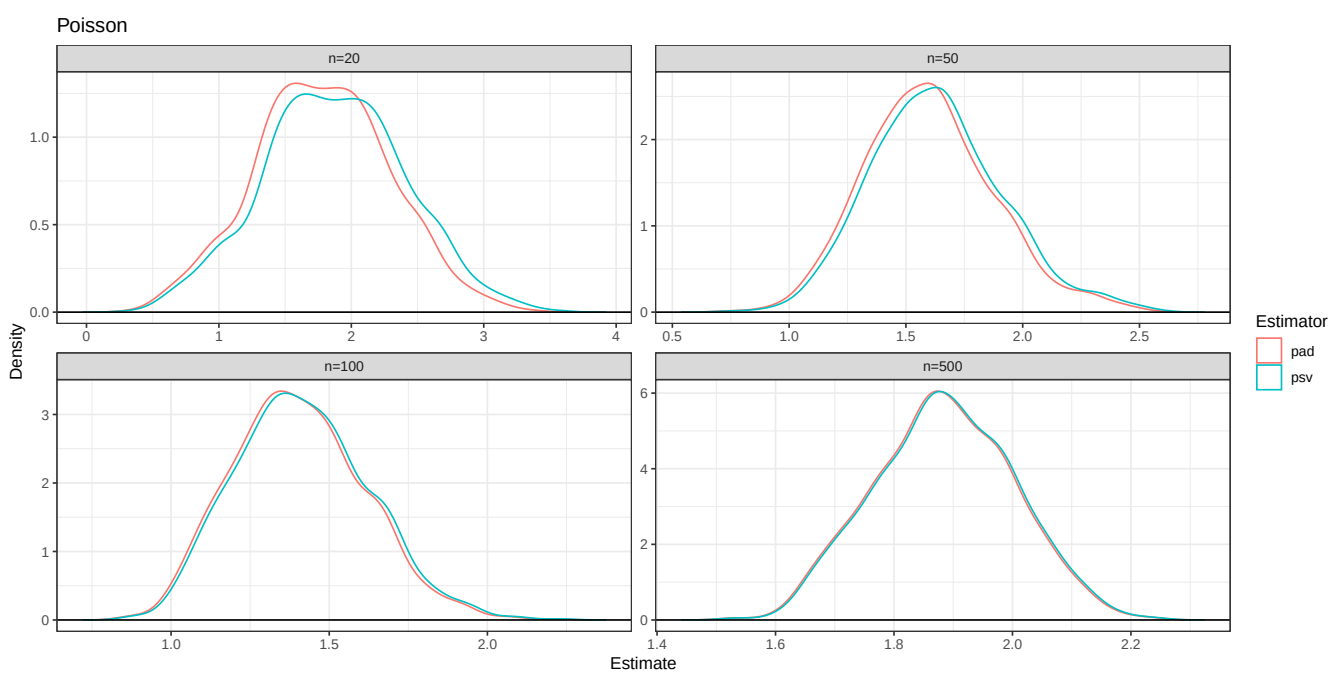
```
set.seed(3434)

n.values = c(20, 50, 100, 500)

# poisson
 pois.sample = list()
 psv_b = list()
 pad_b = list()
 psv = rep(NA, 1000)
 pad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  pois.sample[[as.character(n)]] = rpois(n, 2)
  for(i in 1:1000){
    curr.sample = sample(pois.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    psv[i] = sum((curr.sample - x)^2) / (n - 1)
    pad[i] = sum((curr.sample - x)^2) / n
  }
  psv_b[[as.character(n)]] = psv
  pad_b[[as.character(n)]] = pad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# binomial
 binom.sample = list()
 bsv_b = list()
 bad_b = list()
```

```r
 bsv = rep(NA, 1000)
 bad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  binom.sample[[as.character(n)]] = rbinom(n, 50, (5-sqrt(21))/10)
  for(i in 1:1000){
    curr.sample = sample(binom.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    bsv[i] = sum((curr.sample - x)^2) / (n - 1)
    bad[i] = sum((curr.sample - x)^2) / n
  }
  bsv_b[[as.character(n)]] = bsv
  bad_b[[as.character(n)]] = bad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# normal
 norm.sample = list()
 nsv_b = list()
 nad_b = list()
 nsv = rep(NA, 1000)
 nad = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  norm.sample[[as.character(n)]] = rnorm(n, 0, sqrt(2))
  for(i in 1:1000){
    curr.sample = sample(norm.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
    nsv[i] = sum((curr.sample - x)^2) / (n - 1)
    nad[i] = sum((curr.sample - x)^2) / n
  }
  nsv_b[[as.character(n)]] = nsv
  nad_b[[as.character(n)]] = nad
}
# creates random sample that is sampled 1000 times
#calculates the estimates

# exponential
 exp.sample = list()
 esv_b = list()
 ead_b = list()
 esv = rep(NA, 1000)
 ead = rep(NA, 1000)
 #creates storage for data

for(n in n.values){
# take 1 random sample of different sizes
  exp.sample[[as.character(n)]] = rexp(n, 1/(sqrt(2)))
  for(i in 1:1000){
    curr.sample = sample(exp.sample[[as.character(n)]], size = n, replace = TRUE)
    x = mean(curr.sample)
```

```
    esv[i] = sum((curr.sample - x)^2) / (n - 1)
    ead[i] = sum((curr.sample - x)^2) / n
  }
  esv_b[[as.character(n)]] = esv
  ead_b[[as.character(n)]] = ead
}
# creates random sample that is sampled 1000 times
#calculates the estimates
```

```
ggdat.pb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(psv_b[["20"]], pad_b[["20"]],
psv_b[["50"]], pad_b[["50"]],
psv_b[["100"]], pad_b[["100"]],
psv_b[["500"]], pad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.pb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Poisson") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

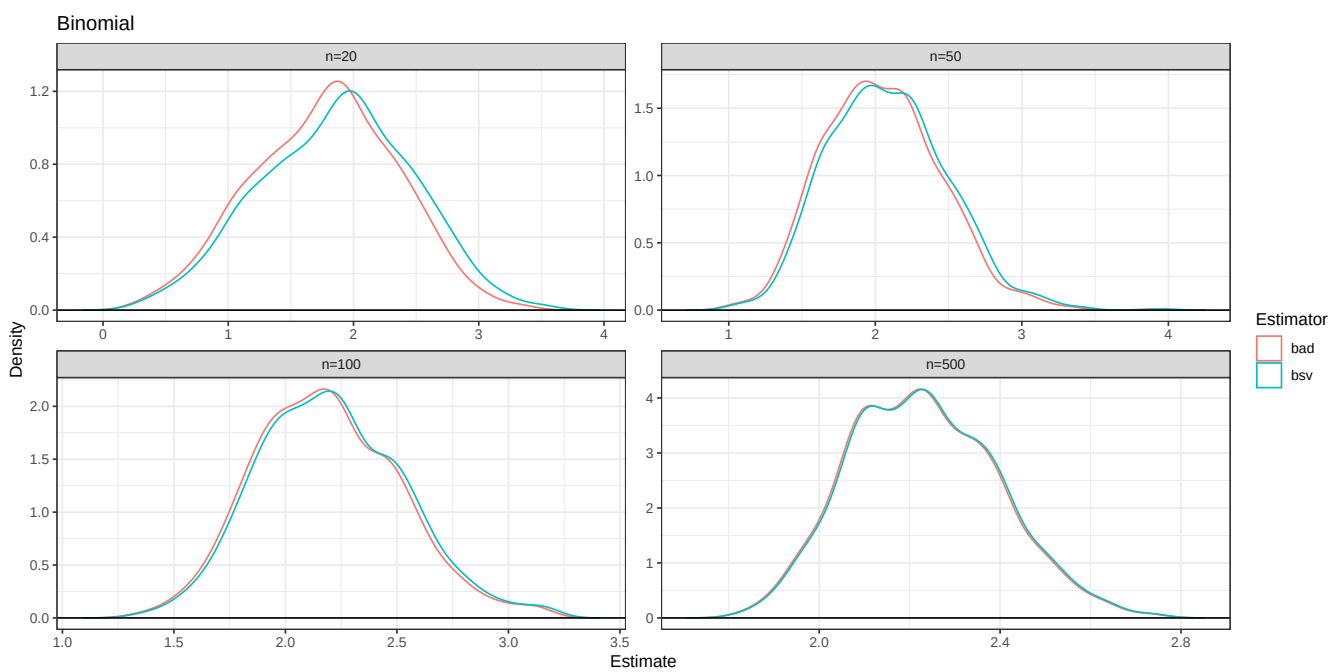
Figure 21: Poisson Density Plot Variance Estimation with Bootstrapping.




```
ggdat.bb <- tibble(
Estimator = rep(c("bsv", "bad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
```

```
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(bsv_b[["20"]], bad_b[["20"]],
bsv_b[["50"]], bad_b[["50"]],
bsv_b[["100"]], bad_b[["100"]],
bsv_b[["500"]], bad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.bb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Binomial") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

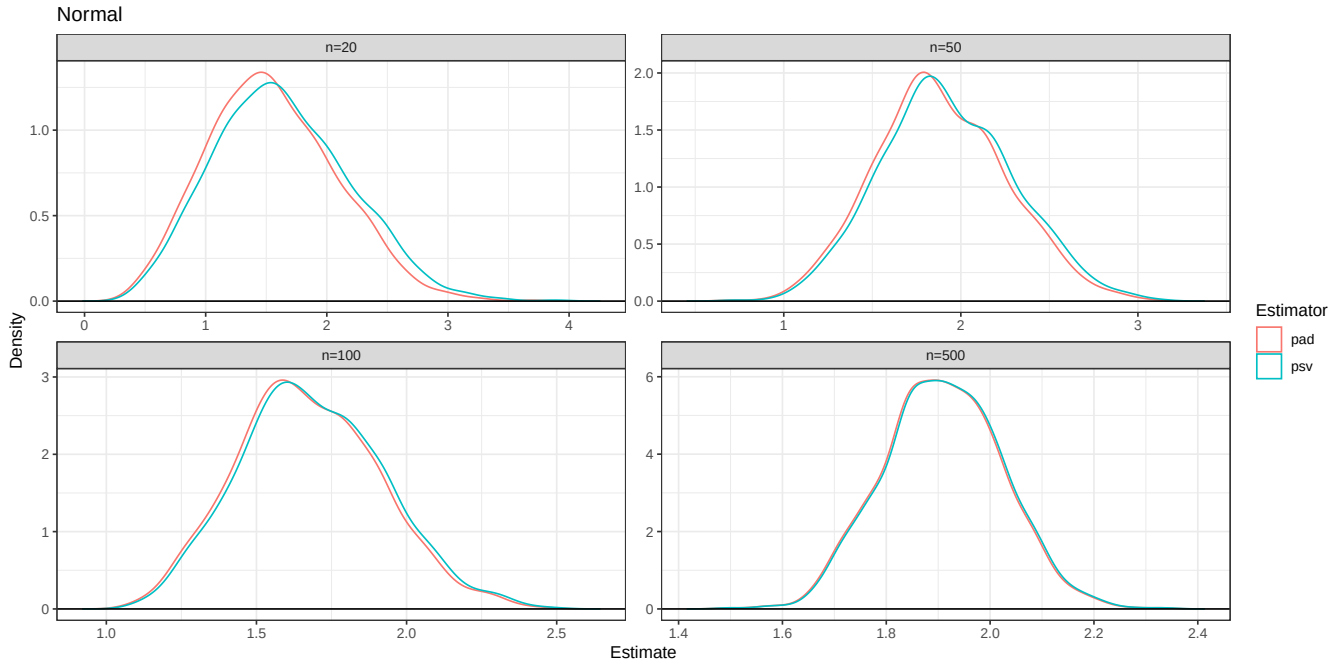
Figure 22: Binomial Density Plot Variance Estimation with Bootstrapping.




```
ggdat.nb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(nsv_b[["20"]], nad_b[["20"]],
nsv_b[["50"]], nad_b[["50"]],
nsv_b[["100"]], nad_b[["100"]],
nsv_b[["500"]], nad_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.nb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Normal") +
```

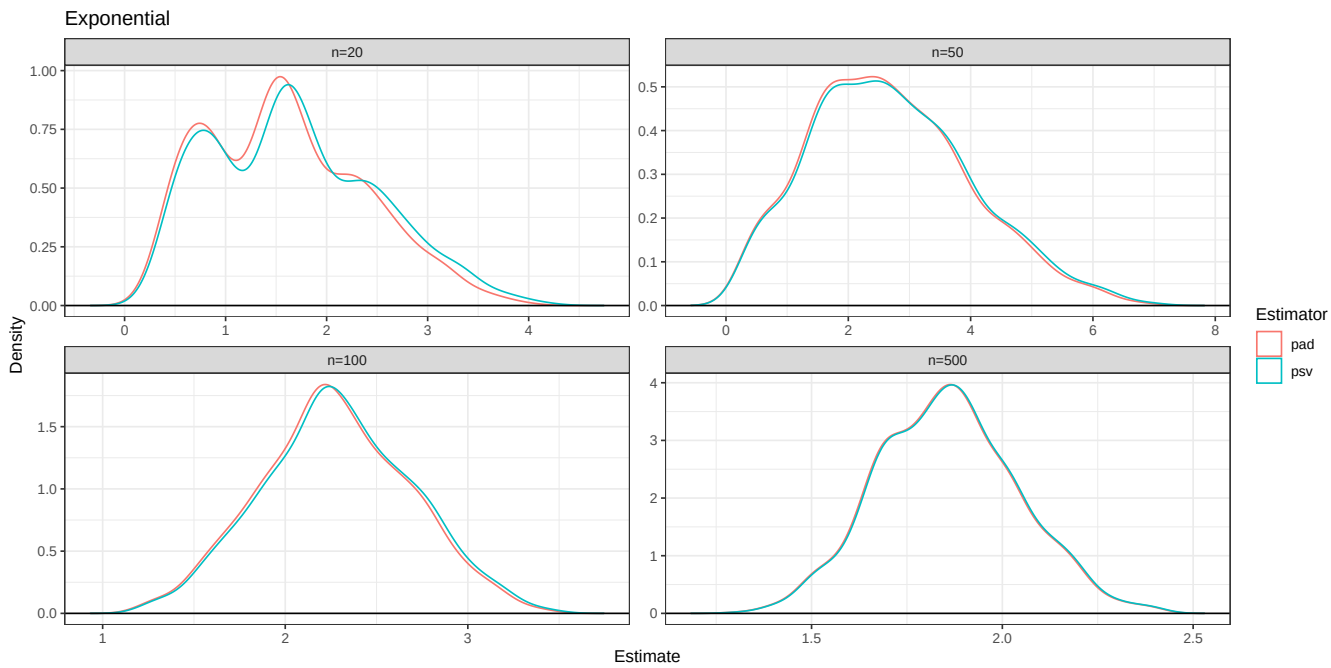
```r
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 23: Normal Density Plot Variance Estimation with Bootstrapping.




```r
set.seed(7272)

ggdat.eb <- tibble(
Estimator = rep(c("psv", "pad"), each = 1000, times = 4),
n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
levels = c("n=20", "n=50", "n=100", "n=500")),
Estimate = c(esv_b[["20"]], ead_b[["20"]],
esv_b[["50"]], ead_b[["50"]],
esv_b[["100"]], ead_b[["100"]],
esv_b[["500"]], ead_b[["500"]])
)
#creates a data set with each n of each estimate

ggplot(ggdat.eb) +
geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
geom_hline(yintercept = 0) +
ggtitle("Exponential") +
# each facet has a different scale
facet_wrap(~ n, scales = "free") +
ylab("Density") +
theme_bw()
```

Figure 24: Exponential Density Plot Variance Estimation with Bootstrapping.



3 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) summary of the simulation and the results using your work in Lab 11. Your summary should be professional, clear, and all claims should be corroborated with statistical support.

By using histograms as my main visualization of data, I was more able to see the potential skew of sampling distributions and the amount of kurtosus of these distributions, however I did have to rely more on my numerical summaries of tables to gain greater insight on the distributions' accuracy. The current sample variance calculation is a better estimate of variance as opposed to the average squared distance from the mean as sampling distributions of these measures of variance demonstrate that the current sample variance measurement as it is often less bias across the poisson, binomial, normal, and exponential distributions. While the current calculation has worse precision, the accuracy is more helpful in estimating a population's true variance. Additionally, the sampling distributions of the sample variance is much less skewed than the average squared distance from the mean, which is often right-skewed, demonstrating that this parameter estimation often favors a lower population variance than the true parameter. The density plots demonstrate that while tbe average squared distance away from the mean is very precise, this estimate is incredibly likely to report a lower variance than the true parameter, while the sample variance is about equally likely to report lower or higher.

In the bootstrapping portion of the lab, it becomes clear that with one sample, the estimates become much similar as there is a lower amount of variability when all the data comes from one sample taken with replacement. This section also demonstrated the importance of a representative sample. When a sample was less representative, the estimates were about equivalent to each other and were centered more towards the initial sample's mean rather than the population mean. This is especially clear in Figure 13 and **?@fig-fig-binomial-bootstrap**, among many other as this became a common trend. Typically both estimates operated similarly, only really deviating from each other when the number of replacements was lower. When the number of replacements is 500, the density plots of both estimates were basically overlapping. This demonstrates that bootstrapping and more limited data that could possibly be less representative makes bias differences between these estimates less prominent.

4 Citations

I used the tidyverse package (Wickham et al. 2019) to read the csv, and both the tidyverse package (Wickham et al. 2019) and the patchwork package (Pedersen 2025) to create graphs, the knitr package (Xie 2025) to help organize

and reference graphs, the ggridges package (Wilke 2025) to help create the density plots, all while using R (R Core Team 2025) as my coding language.

References

Pedersen, Thomas Lin. 2025. *Patchwork: The Composer of Plots*. https://doi.org/10.32614/CRAN.package.patchwork.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

Wilke, Claus O. 2025. *Ggridges: Ridgeline Plots in 'Ggplot2'*. https://doi.org/10.32614/CRAN.package.ggridges.

Xie, Yihui. 2025. *Knitr: A General-Purpose Package for Dynamic Report Generation in R*. https://yihui.org/knitr/.